

Dr. Jim D. Pham

Executive Academic & Technical Leadership Biography

San Jose, California | Adjunct Professor | Software Architect & Educational Researcher

EXECUTIVE SUMMARY & BIOGRAPHY

Dr. Jim Pham is a visionary educational researcher, software architect, and development engineer with extensive expertise spanning computer hardware, software design, and modern technology frameworks. A long-tenured Adjunct Professor of Computer Science, CNET, and Cybersecurity at Ohlone College, he bridges advanced industry innovation with cutting-edge academic pedagogy, driving curriculum evolution and student success. His multifaceted career has had an impact as an author, entrepreneur, doctoral scholar, and community leader, complemented by distinguished engineering contributions at leading technology organizations.

Specialized in technology-driven learning models, agent-based artificial intelligence (AI), cloud architecture, enterprise solutions, and security paradigms, Dr. Pham pioneered the Train of Thought Synthesis (TOTS) Learning Model. Grounded in his doctoral research exploring mobile devices and technology as cognitive mindtools, TOTS provides structured cognitive scaffolding to help students master complex software architectures, sorting algorithms, and system design. He holds a Doctorate in Education with a specialization in Technology and Learning, a Master of Science in Software Engineering, and a Bachelor of Science in Computer Engineering, complemented by advanced certifications in AI, Data Science, and Cybersecurity.

Throughout his industry career, Dr. Pham has led critical engineering initiatives, including VM build execution frameworks and security modules for Oracle Visual Builder Studio, as well as core cross-database migration tools for PeopleSoft. His leadership in collaborative research—notably directing student teams on NASA/CREST satellite data processing systems—earned him the prestigious NASA Ames Honor Award alongside campus recognitions such as the Ohlone Shining Star Award. Beyond his professional and academic pursuits, Dr. Pham remains deeply committed to public service as a certified member of the City of San Jose Community Emergency Response Team (CERT) and Incident Command Systems, as well as an active member of local pastoral leadership councils.

CORE EXPERTISE & PEDAGOGICAL INNOVATION

Pedagogical Innovation	Train of Thought Synthesis (TOTS) Model, Cognitive Scaffolding, Agentic AI in Education, MindTools Framework, LMS Optimization (Canvas, Blackboard).
Cloud & Software Engineering	Oracle Cloud Infrastructure (OCI), Visual Builder Studio (VBS), Microservices Architecture, Java EE, C/C++, Python, SQL Performance Tuning, Data Pipelines.
Security & Community Leadership	Cybersecurity & CCNA Security, Cyber Threat Management, Incident Command Systems (CSTI), San Jose CERT Leadership, Emergency Response.